

Adarsh Bajpai

📞: +91-9460185619 | ✉: abjpr.ind@gmail.com | 🌐: adarshbajpai.com
🌐: linkedin.com/in/adarshbajpai11 | 🐙: github.com/nameadarsh

SUMMARY

AI Engineer building real-world machine learning systems and pipelines from development to deployment, with a focus on performance, reliability, and practical impact.

EXPERIENCE

AI Engineer

Jan 2026 – Present

Global Infoventures Pvt. Ltd.

Delhi, India

- Developed a RAG pipeline for document querying and question answering with support for Hindi text; used chunking strategies to preserve context and improve response quality.
- Contributed to the development of a scalable Learning Management System (LMS) as part of a joint project with TCS, to serve a student base of **2 lakh users** across multiple universities.

PROJECTS

Leaf Disease Detection System [\[GitHub\]](#) | Python, U2-Net, ResNet, Flask, React, OpenCV, NLP

- Built a hybrid deep learning model (**U2-Net + ResNet**) for crop disease detection and severity estimation, achieving **96% classification accuracy** and reducing misclassification rate by **~15%**.
- Extended into a full-stack web app (**React + Flask**) with image upload, bilingual results, severity scoring, and an **LLM chatbot** for real-time farmer queries.

Real-Time Multi-Object Tracking Pipeline [\[GitHub\]](#) | RT-DETR, ByteTrack, OpenCV, Pandas, PyYAML

- Built an end-to-end tracking pipeline using **RT-DETR-R50** and **ByteTrack**, designed for **Indian road environments**; detects and tracks **vehicles, pedestrians, and animals** with stable persistent IDs across video streams.
- Processes videos in batch mode, **producing annotated outputs**, side-by-side comparisons, and per-detection **CSV logs with confidence scores**. Pipeline behavior is configurable via YAML (tracking thresholds, parameters, I/O settings).

Clarity – Modern Guidance, Grounded in Wisdom [\[GitHub\]](#) [\[Live App\]](#) | FastAPI, React, FAISS, RAG, LLMs, TypeScript

- Built a multilingual **RAG-based guidance platform** that retrieves and reranks Bhagavad Gita verses using **Top-K semantic search** to deliver contextually relevant answers for modern-life queries.
- Developed an end-to-end AI pipeline where retrieved verses are processed through **LLM-based reasoning** to generate grounded responses with **real-time TTS speech output** across **11 regional languages**.
- Improved scalability and response reliability by serving static scripture data from the frontend and engineering an async **FastAPI + SSE streaming** backend with background warmup orchestration.

TECHNICAL SKILLS

Programming: Python, JavaScript

Data Science & ML Libraries: PyTorch, NumPy, Pandas, Matplotlib, OpenCV

AI & Machine Learning: Machine Learning, CNNs, ResNet, U2-Net, Vision Transformers, RT-DETR, ByteTrack, Natural Language Processing, RAG Pipelines, LLM Integration, Data Preprocessing, Feature Engineering

Web Frameworks & Development: Flask, FastAPI, React, Node.js, MongoDB

DevOps & Tools: Git, GitHub, Docker, Jenkins, Postman, Jira

Design & Visualization: Figma, Wireframing, UI Design, Prototyping, Adobe Suite

EDUCATION

Bachelor of Technology in Computer Science

Aug. 2022 – May 2026

NIIT University | AI and ML Specialisation

Neemrana, Rajasthan

- Coursework: Machine Learning, Generative AI, AI Agents and Systems, Software Engineering, DBMS

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Generative AI Professional [\[Certificate\]](#)

Oracle | Generative AI, model deployment on OCI, prompt engineering, enterprise AI integration

Model Context Protocol (MCP) Advanced Development Certificate [\[Certificate\]](#)

Anthropic | MCP architecture, StreamableHTTP/SSE transport, sampling, notifications, file access control, deployment and scaling concepts